NIDAR RescueSwarm — Bill of Materials

Three-aircraft programme · Revision 2026-09-19 · all prices INR

Parts, three aircraft	**4,79,628**	Parts per aircraft	**1,34,400**
Released (tax where owed + 15 % contingency)	**5,54,483**	Line items	**57**

Architecture

Flight controller	Holybro Kakute H7 V1.5 (ArduPilot), stacked on a 4-in-1 ESC
Companion	Raspberry Pi 5 4 GB + AI HAT+ 26 TOPS; Arducam 12.3 MP, 6 mm lens
Positioning	ZED-F9P RTK on every aircraft (GenX F9P Air, compass built in); local F9P base, RTCM over E22 LoRa, no NTRIP
Video	Digital, wfb-ng on Pi OS (BL-M8812EU2), CV overlay on the feed; one ground receiver per aircraft
Links	ELRS 2.4 GHz safety pilot (TX12 MKII); E22 868 MHz primarily for RTCM
Power	6S LiPo; ESC current sensor to FC; 8 A UBEC for Pi, 3 A UBEC for video radio, separate servo UBEC

Summary by group

Group	Lines	Per aircraft	Programme
Test instruments	6	–	3,246
Avionics, sensing and RTK	21	72,317	2,44,077
Airframe, propulsion and power	19	62,083	1,93,916
Flight-test support	7	–	29,867
Ground station	4	–	8,522
Total parts	**57**	**1,34,400**	**4,79,628**

Line items

Item	Part	/AC	Qty	Unit	Total	Source
Test instruments						**3,246**
Load-cell amplifier	7Semi HX711 24-bit ADC	–	1	104	104	Robu
Load cell	REES52 20 kg load cell + HX711 breakout	–	1	854	854	Amazon India
Thrust-stand mast	SURYAKUSH metal 2.1 m tripod	–	1	373	373	Flipkart
Fasteners	Hex socket-head assortment M3/M4/M5, SS304	–	1	1,140	1,140	OnlyScrews
Threadlocker	Loctite 243, 50 ml	–	1	664	664	Amazon India
Throttle source	Servo tester / ESC consistency tester, PPM	–	1	111	111	Robokits
Avionics, sensing and RTK						**2,44,077**
Flight controller	Holybro Kakute H7 V1.5, no Bluetooth	1	3	9,269	27,807	Indian Robo Store
FC vibration mount	Glass-fibre anti-vibration shock absorber	1	3	161	483	Robu
Autopilot log card	SanDisk 32 GB microSDHC, class 10	1	3	1,549	4,647	Robocraze
Companion computer	Raspberry Pi 5, 4 GB	1	3	11,049	33,147	Yantronix
AI accelerator	Raspberry Pi AI HAT+, 26 TOPS	1	3	11,749	35,247	Robu
Compute cooling	Official Raspberry Pi 5 active cooler	1	3	519	1,557	Robu
Storage	SanDisk High Endurance 128 GB microSD	1	3	3,989	11,967	Flipkart
GNSS RTK rover	GenX F9P Air: ZED-F9P + helix + compass, ∼20 g	1	3	15,392	46,176	Robokits
GNSS RTK base	SmartElex ZED-F9P-04B RTK board, U.FL	–	1	17,965	17,965	Robocraze
GNSS antenna, base	LS-125-A L1/L2/L5 survey antenna	–	1	8,123	8,123	Robokits
GNSS feeder, base	U.FL to SMA female pigtail, 10 cm	–	1	49	49	Robocraze
RTK base radio	EByte E22-900T22D-V2, SX1262, 22 dBm	–	1	989	989	Robu
Height rangefinder	Benewake TFmini-S, 12 m lidar, UART	1	3	2,827	8,481	Indian Robo Store
Camera + lens	Arducam 12.3 MP 1/2.3 in HQ module, 6 mm CS lens	1	3	6,799	20,397	Robu
Video transmitter	LB-Link BL-M8812EU2, RTL8812EU, 5 GHz 2T2R	1	3	1,849	5,547	Hubtronics
Video antennas	6 dBi 5 GHz dipole + 165 mm U.FL lead	2	6	75	450	Robocraze
Command receiver	ExpressLRS RX24T, 2.4 GHz	1	3	1,439	4,317	Robu
Coordination radio	EByte E22-900T22D-V2, SX1262, 22 dBm	1	3	989	2,967	Robu
BEC, primary	8 A UBEC, 5/6 V out, 2-6S in	1	3	3,939	11,817	Amazon India
BEC, video radio	Hobbywing 3 A UBEC, 5/6 V, 5 A peak, 2-6S	1	3	369	1,107	KitsGuru
BEC, secondary	ReadytoSky 2-6S 5V/3A + 12V/3A switchable UBEC	1	3	279	837	Robu
Airframe, propulsion and power						**1,93,916**
Motors	Tarot TL96020, 5008, 340 KV	4	12	4,703	56,436	Robokits
Propellers	1855 counter-rotating carbon fibre, CW+CCW pair	4	16	1,641	26,256	Robokits
Speed controllers	Hobbywing XRotor FPV G2, 4-in-1, 65 A, 3-6S	1	3	6,500	19,500	RC Mumbai
Arm tube	3K carbon fibre, OD25 x ID23 x 1000 mm	2	6	2,259	13,554	Robu
Frame plate stock	3K carbon fibre sheet, 300 x 200 x 2 mm	1	3	2,895	8,685	Indian Robo Store
Motor mounts	Tarot 25 mm motor mount, TL9602	4	12	1,119	13,428	Robu

continued

Item	Part	/AC	Qty	Unit	Total	Source
Arm clamps	T-bolt hose clamps 23-25 mm, SS304, 4 pcs	1	3	3,012	9,036	Amazon India
Landing gear	F450/F550 landing skid	1	3	549	1,647	Robu
Suspension springs	5 mm stainless double torsion spring	10	32	2	64	IndiaMART
Printed parts	Pro-Range PETG-CF filament, 1.75 mm, 1 kg	–	1	949	949	Robu
Release servos	MG90S mini servo, 180 deg	4	12	149	1,788	Robocraze
Battery pack	GenX 22.2V 6S 12000mAh 25C/50C LiPo, XT90	1	3	9,088	27,264	Robokits
Pack fusing	ANL fuse holder + 150 A ANL fuses	1	3	3,000	9,000	Njour
Pack retention	300 mm LiPo strap, reusable	2	6	62	372	Robu
Main leads	10 AWG ultra-flexible silicone wire	–	1	149	149	Robu
Power connectors	Amass XT90S anti-spark, M/F pair	4	12	180	2,160	Robokits
Signal connectors	JST ZH 6-pin, 200 mm leads	1	3	23	69	Robu
Antenna feeders	SMA M-F bulkhead RG316, 150 mm + adapters	1	3	1,139	3,417	Desertcart
Insulation	PVC heat-shrink sleeve, 150 mm	1	3	47	141	Robu
Flight-test support						**29,867**
Safety-pilot transmitter	RadioMaster TX12 MKII ELRS + RP1 receiver	–	1	16,519	16,519	zBotic
Battery charger	ToolKitRC M6D, 500 W, 25 A, dual	–	1	7,349	7,349	Robu
Cable management	Nylon zip ties, 350 mm, 100 pcs	–	1	135	135	Robu
Heat-shrink kit	328 pcs assorted 2:1 heat-shrink	–	1	159	159	Robu
Mounting tape	3M VHB 5952, 6 mm x 8.2 m	–	1	560	560	Amazon India
Hook and loop	Self-adhesive hook and loop, 25 mm x 2 m	–	1	145	145	Amazon India
Consumables	Fasteners, tape, connectors, filament	–	1	5,000	5,000	local
Ground station						**8,522**
Video receivers	LB-Link BL-M8812EU2, RTL8812EU, one per aircraft channel	–	3	1,849	5,547	Hubtronics
Receive antennas	6 dBi 5 GHz dipole + U.FL lead, 2 per receiver	–	6	75	450	Robocraze
Coordination base	EByte E22-900T22U, SX1262, USB	–	1	1,646	1,646	Hubtronics
Base station mount	Photographic tripod and adapter	–	1	879	879	Amazon India
Total parts, three aircraft						**4,79,628**

/AC = fitted per aircraft (– = bought once for the programme). *Qty* = programme quantity, including spares. Retail listings are GST-inclusive; GST is added only to quotation and B2B lines (14,064 of parts). Every source name links to the listing the price came from.

Open before ordering

- **LiPo capacity.** The 12 Ah pack is a placeholder. The landing-queue reserve needs $\geq$ 287 Wh ($\approx$ 13 Ah at 6S), which it misses. Size the pack from the final AUW and hover current.
- **Failsafe parameters.** `params.py` and the pack model still describe the Li-ion pack. Update them with the LiPo, and before flying it.
- **ESC.** 65 A vs 50 A is decided on the thrust stand: desync, current and temperature on 18 in props with 340 KV motors.
- **Confirm in the cart:** the GenX F9P Air's bands (L1/L2), the LiPo's capacity (10 vs 12 Ah on the listing), and the survey antenna's connector.
- **Not yet in this BOM:** the ground directional antenna and a powered USB hub for the three video receivers.